



GORDON COLLEGE COLLEGE OF COMPUTER STUDIES

OLONGAPO CITY

• BS in Computer Science • BS in Entertainment and Multimedia Computing - Digital Animation Technology
• BS in Entertainment and Multimedia Computing - Game Development • BS in Information Technology

A Deployment Framework for an IoT-Based QR Code Attendance and Student Information Management System with AI and Cloud Technologies

1. Technical Architecture & Stack

The system follows a **Cloud-Edge Hybrid** architecture. The IoT device handles high-speed scanning locally, while the cloud manages the database and AI processing.

Layer	Technology	Role in the System
Edge Hardware	ESP32-CAM / Raspberry Pi	Captures QR codes and performs initial validation.
IoT Protocol	MQTT / HTTPS	Transmits logs from the hardware to the server.
Frontend	Next.js 15 (PWA)	Admin dashboard for real-time monitoring.
Backend	Node.js (NestJS) or FastAPI	Manages student records and IoT device registration.
Database	MongoDB / Supabase	Stores time-series attendance logs and student bio-data.
AI Layer	TensorFlow.js / Scikit-learn	Analyzes patterns to predict truancy or anomalies.

2. Core Functional Modules

Module A: IoT Gateway & QR Scanning

- **Feature 1: Secure QR Generation:** The system generates encrypted QR codes for students to prevent unauthorized duplication.
- **Feature 2: Real-time Capture:** The IoT camera decodes QR data and checks for a valid "ID" and "Timestamp."
- **Feature 3: Edge Feedback:** The IoT station displays a "Welcome [Name]" message on an LCD or triggers a buzzer for success.

Module B: Student Information Management (SIMS)

- **Feature 1: Digital Student Profiles:** Stores LRN, contact details, medical info, and emergency contacts.
- **Feature 2: Enrollment Management:** Form for bulk student registration via CSV or individual entry.
- **Feature 3: Family Link:** Associates student accounts with parent contact details for automated SMS/Email alerts.

Module C: AI-Driven Attendance Analytics

- **Feature 1: Anomaly Detection:** AI identifies "Buddy Punching" (scanning for a friend) by analyzing time-gap patterns.



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- **Feature 2: Truancy Forecasting:** Predicts students likely to drop out based on a "3-day absence" trend or frequent tardiness.
- **Feature 3: Peak-Hour Analysis:** Visualizes "In-and-Out" traffic to help school security optimize gate flow.

Module D: Cloud Reporting & Notification

- **Feature 1: Real-time Live Log:** A live feed on the dashboard showing every scan as it happens.
- **Feature 2: Automated Alerts:** Triggers a cloud-based SMS service (like Twilio) to notify parents when a student enters or leaves the campus.
- **Feature 3: Attendance Heatmaps:** Monthly visual reports showing attendance percentages per section.

3. Interface Design Outline (Sitemap)

1. Admin Dashboard

- **Live Metrics:** Total Present Today, Late Arrivals, IoT Device Status (Online/Offline).
- **Alert Feed:** Urgent flags for students missing for 3+ days.
- **IoT Status:** Health check for all scanning stations at different school gates.

2. Student Management (Submenu)

- **[Form] Student Registration:**
 - Fields: Name, LRN, Grade/Section, Parent Mobile, Photo Upload.
- **[Form] QR Generator:** A tool to batch-print ID cards with pre-generated QR codes.

3. Attendance Logs

- **Searchable History:** Filter logs by Date, Section, or Student Name.
- **[Form] Manual Correction:** Pop-up for teachers to manually mark a student as "Excused" or "Field Trip."

4. AI Insights Hub

- **Truancy Map:** List of students flagged by the AI as "High Risk of Dropping Out."
- **Trend Graphs:** Weekly attendance vs. Monthly averages.

5. Hardware Configuration

- **[Form] Register New Device:** Add MAC address/Device ID for new IoT scanners.
- **Security Settings:** API key management for the IoT-to-Cloud bridge.

4. System Flow Guide (The Developer's Logic)



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1. **Phase 1 (Hardware Interaction):** Student places QR code in front of the IoT camera.
2. **Phase 2 (Edge Validation):** The IoT device checks if the QR is in the correct format. It blinks **Green** for local success.
3. **Phase 3 (Cloud Transmission):** The device sends a JSON payload {student_id, gate_id, timestamp} to the cloud via MQTT.
4. **Phase 4 (AI Processing):** The server updates the student's status and runs the anomaly model (e.g., "Has this student already scanned in today?").
5. **Phase 5 (Notification):** The Cloud triggers an SMS to the parent: *"Your child, [Name], entered the school at 7:30 AM."*
6. **Phase 6 (Reporting):** The Admin views the live update on the Dashboard.

5. Security & Privacy (Deployment Framework)

- **IoT Security:** All hardware communication is encrypted using **TLS/SSL** to prevent "man-in-the-middle" attacks on student data.
- **Cloud Privacy:** Personal Identifiable Information (PII) is encrypted at rest using **AES-256**.
- **Physical Safety:** The IoT enclosure must be tamper-proof and weather-resistant (IP65) if placed at outdoor gates.



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